

DYNAMICAL SYSTEMS - MA3081

Exam

Answers always require detailed justification e.g. in the form of deductions, calculations or clear references to results from the lecture. When referring to results from the lecture, state and check its assumptions.

You can answer the questions in any order, but always indicate which question you are working on. Answers to one question should not be split into multiple pieces, but written down coherently. Do not use the color red.

This examination consists of 6 questions.

Problem 1 (9 points). For each of the following differential equations determine the (local) stability of the equilibrium at the origin; in the case of stability, comment on asymptotic stability.

(a) $\begin{pmatrix} x \\ y \end{pmatrix}' = \begin{pmatrix} -3x + \sin(y) \\ -2y \end{pmatrix}$

(b) $\begin{pmatrix} x \\ y \end{pmatrix}' = \begin{pmatrix} -x^3 + y \\ -x - y^5 \end{pmatrix}$

(c) $x' = \alpha x^3 + \beta x^4$, depending on the parameters $\alpha, \beta \in \mathbb{R}$

Solution:

- (a) The linearization at the origin is given by $\begin{pmatrix} -3 & 1 \\ 0 & -2 \end{pmatrix}$, which has eigenvalues -3 and -2 . Consequently $d_s = 2$, $d_u = 0$, $d_c = 0$.

Since the equilibrium is hyperbolic and all eigenvalues are stable, by Hartman-Grobman Theorem we can conclude that the origin is asymptotically stable.

- (b) The linearization at the origin is given by $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ which has eigenvalues $-i$ and i . Consequently $d_s = 0$, $d_u = 0$, $d_c = 2$. **Note: Since the equilibrium is non-hyperbolic, the linearization is inconclusive regarding stability!**

We use the Lyapunov function $L(x, y) = x^2 + y^2$. We have $L(x, y) > 0$ for all $(x, y) \neq (0, 0)$ as well as $L(0, 0) = 0$. Furthermore,

$$\begin{aligned} \langle \nabla L(x, y), f(x, y) \rangle &= \left\langle \begin{pmatrix} 2x \\ 2y \end{pmatrix}, \begin{pmatrix} -x^3 + y \\ -x - y^5 \end{pmatrix} \right\rangle \\ &= -2x^4 + 2xy - 2xy - 2y^6 = -2x^4 - 2y^6 < 0 \end{aligned}$$

for all $(x, y) \neq (0, 0)$. Hence the origin is asymptotically stable.

(c) The linearization at the origin is 0 for any choice of the parameters $\alpha, \beta \in \mathbb{R}$, so $d_s = 0$, $d_u = 0$, $d_c = 1$. For the stability, note that the cubic term determines the sign of the vector field close to the origin, if $\alpha \neq 0$. Sign considerations or phase portraits of the 1-d system yield the following stability properties:

- $\alpha < 0$: asymptotically stable
- $\alpha > 0$: unstable
- $\alpha = 0$:
 - * $\beta \neq 0$: unstable
 - * $\beta = 0$: stable, but not asymptotically stable (everything is fixed)

Problem 2 (6 points). Provide an example of a dynamical system that is conjugate to the shift on the space of sequences of four symbols. Explain your example in detail and justify the conjugacy.

Solution: For example: (a) adapt the Smale horseshoe construction with two horseshoes/three U-turns one after the other

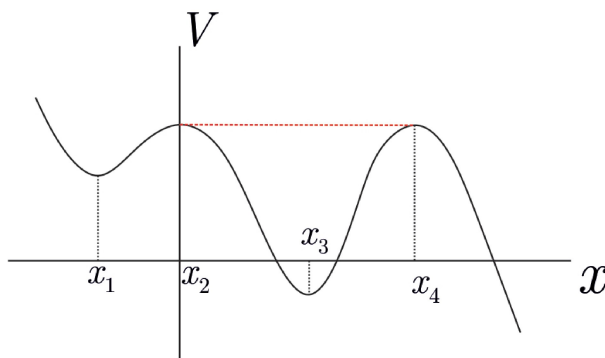
(b) consider the second iterate of the classic horseshoe map

(c) circle quadruple map/second iterate of the circle doubling map: $x \mapsto 4x \bmod 1$

Problem 3 (6 points). Consider a one degree of freedom mechanical system given by

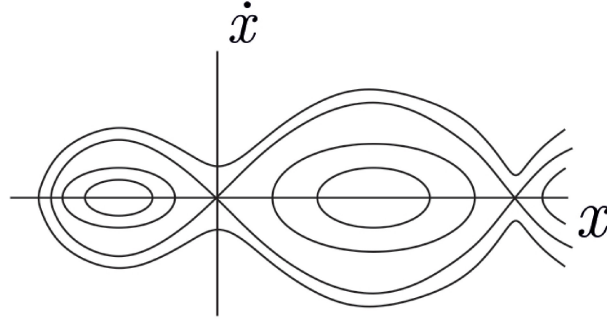
$$\frac{d^2x}{dt^2} + \frac{dV}{dx}(x) = 0,$$

where $V \in C^\infty(\mathbb{R}, \mathbb{R})$ is a smooth potential of the following form:



Draw the corresponding phase portrait in the x - x' -plane.

Solution:



with arrows to the right in the upper half-plane and to the left in the lower half-plane.

Problem 4 (6 points). On the unit circle consider the tripling map $f: [0, 1) \rightarrow [0, 1)$, $x \mapsto 3x \bmod 1$.

- Determine the number of periodic orbits of (minimal) period 2.
- For each 2-periodic orbit specify one point of the orbit.

Solution:

- The second iterate of f is given by $f^2(x) = 9x \bmod 1$. It has 8 fixed points. Furthermore, there are two fixed points of f and these are also fixed points of the second iterate f^2 . The remaining 6 points are of minimal period 2, hence we have 3 orbits of period 2.
- We obtain the 2-periodic points by solving the equation $9x = x + k$ with some $k \in \mathbb{N}$. Solutions are $\frac{k}{8}$, $k \in \{0, 1, \dots, 7\}$. Note that 0 and $\frac{4}{8}$ are also fixed points of f . This gives the following two-periodic orbits:

$$\begin{aligned} \bullet \quad \frac{1}{8} &\mapsto \frac{3}{8} \mapsto \frac{1}{8} \\ \bullet \quad \frac{2}{8} &\mapsto \frac{6}{8} \mapsto \frac{2}{8} \\ \bullet \quad \frac{5}{8} &\mapsto \frac{7}{8} \mapsto \frac{5}{8} \end{aligned}$$

Problem 5 (8 points). We consider the following system, which has a two dimensional center subspace $E^c = \left\langle \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\rangle$ at the origin $(x, y, \lambda) = (0, 0, 0)$:

$$\begin{aligned} x' &= \lambda x - xy, \\ y' &= -y + x^2 + 2\lambda x^2 - 2x^2y, \\ \lambda' &= 0. \end{aligned} \tag{1}$$

- Suppose the center manifold is given as the graph of some function

$$h: \mathbb{R}^2 \rightarrow \mathbb{R}, \quad (x, \lambda) \mapsto a + b_1x + b_2\lambda + c_1x^2 + c_2x\lambda + c_3\lambda^2 + \mathcal{O}(3) =: y,$$

for some constants $a, b_1, b_2, c_1, c_2, c_3 \in \mathbb{R}$. In the given setting, which facts allow to conclude $a = b_1 = b_2 = 0$?

- Show that the set $W = \{(x, y, \lambda) \in \mathbb{R}^3 \mid y = x^2\}$ is invariant under (1).

- (c) Consequently the set W is a center manifold of (1). Determine the bifurcation behaviour of the reduced system on W using λ as the bifurcation parameter and draw the corresponding bifurcation diagram.

Solution:

- (a) Since the center manifold is tangentially to the center space, which is in our case the two dimensional subspace spanned by the x and λ -coordinates, the linear terms b_1x and $b_2\lambda$ vanish. Furthermore the center manifold passes through the origin, so it follows that $a = 0$.
- (b) We verify that $y = h(x, \lambda) = x^2$ satisfies the invariance equation

$$y' = \partial_x h(x, \lambda)x' + \partial_\lambda h(x, \lambda)\lambda'.$$

For our system and with $y = x^2$ this reads as

$$-x^2 + x^2 + 2\lambda x^2 - 2x^4 = 2x(\lambda x - x^3).$$

- (c) The reduced equations are given by

$$\begin{aligned} x' &= \lambda x - x^3 \\ \lambda' &= 0 \end{aligned}$$

which is the normal form of a pitchfork-bifurcation.

Problem 6 (3 + 5 points).

- (a) Determine two ergodic measures of the map

$$f: [0, 1] \rightarrow [0, 1], \quad x \mapsto \frac{5}{3}x \pmod{1}.$$

- (b) Let $(X, \mathcal{B}, \mu)$ be a probability space and $\varphi: X \rightarrow X$ a μ -preserving map. Show that φ is ergodic if and only if for all sets $A, B \in \mathcal{B}$ with $\mu(A) > 0$ and $\mu(B) > 0$ there is a $k \in \mathbb{N}$ such that $\int_A \varphi_*^k(\chi_B) d\mu > 0$.

Solution: (a) From the lecture we know that the Dirac measure on any fixed point or the averaged Dirac measures on a periodic orbit are ergodic measures. Thus, we look for fixed points and periodic orbits, and find the only fixed point 0 and the period-2 orbit $(\frac{9}{16}, \frac{15}{16})$. Thus the measures δ_0 and $\frac{1}{2}(\delta_{\frac{9}{16}} + \delta_{\frac{15}{16}})$ are ergodic measures of f .

Note: the map is expanding, so the Lebesgue measure cannot be invariant!

(b) Note: $\int_A \varphi_*^k(\chi_B) d\mu = \mu(A \cap \varphi^{-n}(B))$, and this equivalence was stated, but not proved, in Theorem 4.10 in the lecture.

“ $\Rightarrow$ ”: Let φ be ergodic and take sets $A, B \in \mathcal{B}$ with $\mu(A), \mu(B) > 0$. By Proposition 4.17 we have

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} \mu(A \cap \varphi^{-k}(B)) = \mu(A)\mu(B) > 0.$$

If now we had $\mu(A \cap \varphi^{-n}(B)) = 0$ for all $k \in \mathbb{N}$, then also the above limit would be zero, giving a contradiction.

“ $\Leftarrow$ ”: Assume that for all sets $A, B \in \mathcal{B}$ with $\mu(A) > 0$ and $\mu(B) > 0$ there is a $k \in \mathbb{N}$ such that $\mu(A \cap \varphi^{-k}(B)) > 0$ but φ is not ergodic. Then there is an invariant set B with $\mu(B) \notin \{0, 1\}$. This means that also $\mu(B^c) \notin \{0, 1\}$. By the above assumption (applied to $A = B^c$) there is some $k \in \mathbb{N}$ with $0 \neq \mu(B^c \cap \varphi^{-k}(B)) = \mu(B^c \cap B) = \mu(\emptyset) = 0$, where we exploited the invariance of B for the second equality. This is clearly a contradiction, so φ must be ergodic.